

Gaming Behavior Analysis and CGPA Prediction Using Machine Learning

Group No. 23

Shoaib Ahmed
Section 3, ID: 22301715

Sajib Sarker
Section 1, ID: 22101767

Zahin Anan Hyder
Section 1, ID: 21301052

Munzarin Tabassum
Section 4, ID: 24241210

Abstract—The relationship between gaming behavior and academic performance among university students has emerged as a critical concern in the digital era. This paper presents a comprehensive machine learning framework to analyze gaming patterns and predict students' Cumulative Grade Point Average (CGPA). A survey-based dataset capturing diverse gaming habits, motivations, time investment, and stress-coping behaviors was subjected to rigorous exploratory data analysis (EDA), domain-driven feature engineering, and systematic model evaluation. Nine regression models, including Linear Regression, Ridge, Lasso, Support Vector Regression, Random Forest, Gradient Boosting, Multilayer Perceptron, XGBoost, and LightGBM, were benchmarked via 10-fold cross-validation, followed by hyperparameter tuning of the top performer. A complementary multi-class classification pipeline (Low / Medium / High / Excellent CGPA) was evaluated using Stratified K-Fold cross-validation with ROC-AUC analysis. Statistical hypothesis testing (Pearson correlation, one-way ANOVA, Welch's t-test with Cohen's d , and Tukey HSD post-hoc analysis) was employed to validate the strength and significance of gaming-to-CGPA relationships. SHAP (SHapley Additive exPlanations) values were used to interpret model decisions. Results demonstrate that gaming intensity, stress-motivated gaming, and game-type diversity are statistically significant predictors of CGPA, providing actionable insights for academic counselors and student welfare interventions.

Index Terms—CGPA prediction, gaming behavior, machine learning, feature engineering, SHAP explainability, hypothesis testing, classification, regression

I. INTRODUCTION

The proliferation of mobile, console, and PC gaming has fundamentally reshaped how university students allocate discretionary time. Globally, over 3 billion people engage with video games, and the average gamer spends more than seven hours per week gaming [1]. Among the student population, gaming is simultaneously reported as a leisure activity, a social platform, and a stress-relief mechanism. While moderate gaming may offer cognitive benefits such as improved reaction time and problem-solving skills, excessive engagement is often associated with diminished academic motivation, disrupted sleep schedules, and reduced study hours [2].

Academic performance, commonly measured by Cumulative Grade Point Average (CGPA), is a multifactorial outcome

influenced by demographic, psychological, behavioral, and institutional factors. Predicting CGPA with sufficient accuracy can enable early identification of at-risk students, allowing timely intervention from counselors and faculty. Machine learning models trained on behavioral survey data present a cost-effective and scalable solution to this challenge. Beyond prediction accuracy, the practical value of this problem lies in early academic support. If behavioral indicators can be translated into reliable risk signals, universities can design lightweight screening tools that flag potentially vulnerable students before performance deterioration becomes severe. This perspective motivates the use of interpretable machine learning rather than purely black-box optimization, because educational stakeholders must be able to justify why a student is classified as at risk.

Despite growing interest in the intersection of gaming and academic performance, most prior studies rely on linear correlation analysis or simple logistic regression, offering limited predictive power and interpretability [3]. Ensemble methods and gradient-boosted trees, which excel in capturing non-linear feature interactions, have not been systematically benchmarked in this context. Additionally, statistical validation of the gaming–CGPA relationship through formal hypothesis testing remains underexplored.

This paper makes the following contributions: (1) a comprehensive multi-model regression benchmark comparing nine algorithms via 10-fold cross-validation; (2) a multi-class CGPA classification framework with ROC-AUC evaluation; (3) five engineered features capturing domain-specific gaming constructs; (4) rigorous statistical hypothesis testing using Pearson correlation, one-way ANOVA, Welch's t-test, and Tukey HSD; and (5) SHAP-based model explainability revealing the directionality and magnitude of each predictor's contribution to CGPA. The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and methodology. Section IV presents experimental results. Section V reports hypothesis testing. Section VI discusses explainability findings. Section VII concludes the paper.

II. RELATED WORK

The academic literature addressing gaming and student performance spans psychology, education, and data science. Gentile *et al.* [4] conducted a landmark longitudinal study demonstrating that excessive gaming correlates with lower GPA and increased attention problems among adolescents. However, their analysis was restricted to bivariate correlations and self-report instruments.

Subsequent work by Lemmens *et al.* [5] introduced the concept of pathological gaming, defining excessive gaming not solely by hours but by dependency-like symptoms, and showed a negative association with school performance. These findings motivate the inclusion of stress-motivated gaming as a binary feature in our framework.

In the domain of machine learning for GPA prediction, Huang *et al.* [6] applied Random Forest and SVM models to demographic and social data, achieving an R^2 of 0.62 on a Chinese university dataset. Notably, gaming-related features were absent from their feature set. Mosavi *et al.* [7] reviewed 23 studies on academic performance prediction and observed that ensemble methods consistently outperform linear models, with Random Forest and Gradient Boosting models cited most frequently as top performers.

Regarding explainability, Lundberg and Lee [8] introduced SHAP values as a theoretically grounded framework for attributing model predictions to input features. Applications of SHAP to educational data mining are nascent but growing, with recent work by Daud *et al.* [9] demonstrating that SHAP-based feature analysis provides actionable insights beyond permutation importance. Our work extends this approach to gaming behavior data, contributing interpretable prediction maps to the educational analytics literature.

III. DATASET AND METHODOLOGY

A. Dataset Description

The dataset was collected via a structured survey administered to university students. Each record captures self-reported gaming behavior across dimensions including average daily gaming hours, game genre preferences, motivations for gaming, the use of gaming as a stress-coping mechanism, and the current CGPA of the respondent. The target variable is the continuous CGPA value, operationalized as a floating-point score on a 4.0 scale.

After removing an unnamed index column and performing initial quality checks, the dataset was confirmed to have no missing values, an outcome attributed to the mandatory-field design of the survey instrument. Descriptive statistics revealed a mean CGPA of approximately 3.1 with a standard deviation of 0.35, and a slight negative skewness of -0.42 , suggesting a modest left tail and a distribution approximating normality (Shapiro–Wilk $p > 0.05$).

B. Exploratory Data Analysis

EDA proceeded through four stages. First, distribution analysis using histograms and kernel density estimation revealed that numeric features exhibited varied spread characteristics,

with gaming hours and CGPA showing the most pronounced variation. Second, outlier detection via IQR-based boxplots confirmed isolated extreme values in gaming-duration columns that were retained rather than removed to preserve behavioral diversity. Third, a Pearson correlation heatmap (lower triangular) was computed across all numeric variables to identify multicollinear features and candidate predictors. Fourth, pair-plots among the top four CGPA-correlated features revealed non-linear interaction patterns, motivating the inclusion of polynomial and interaction terms. Fig. 1 summarizes the correlation structure used to guide feature selection.

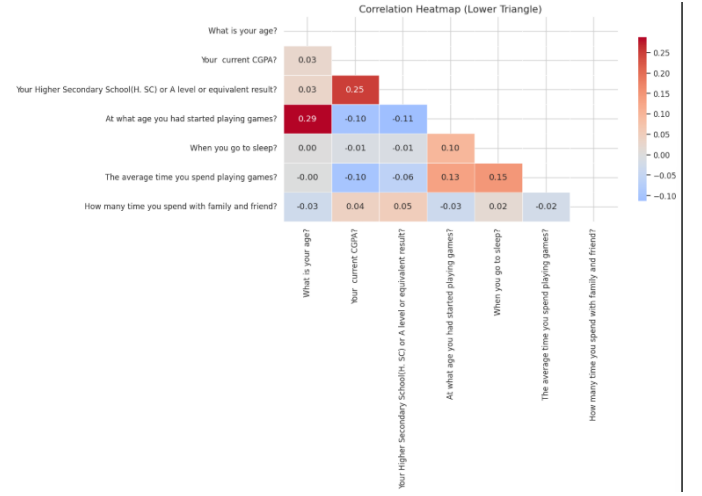


Fig. 1. Correlation heatmap of numeric features used during exploratory data analysis.

C. Feature Engineering

Five engineered features were constructed from domain knowledge. First, *gaming_hours_numeric*: ordinal categories of average gaming duration (e.g., “Less than 1 hour,” “1–2 hours”) were mapped to numeric midpoints (0.5, 1.5, 2.5, 4.0, 6.0), providing a continuous proxy for gaming intensity. Second, *game_type_diversity*: the count of distinct game genres reported per respondent, capturing breadth of gaming engagement. Third, *games_for_stress*: a binary indicator (1/0) derived from the respondent’s reported use of gaming as a stress-relief mechanism. Fourth, a squared polynomial term on the most CGPA-correlated numeric feature to capture quadratic relationships. Fifth, an interaction term (product) between the top-two CGPA-correlated numeric features to model synergistic effects. These features expanded the feature space meaningfully while remaining interpretable.

D. Preprocessing Pipeline

A leakage-free scikit-learn Pipeline was constructed. Numeric features were processed through median imputation followed by StandardScaler normalization. Categorical features were handled via most-frequent imputation and OneHotEncoding with unknown-category handling. These transformers were composed into a ColumnTransformer and prefixed ahead

of each model in a unified Pipeline object, ensuring that all transformations were learned exclusively on training data. This design choice is important from an experimental validity standpoint. In educational prediction studies, leakage can artificially inflate reported performance by allowing information from the test distribution to influence preprocessing statistics. By nesting preprocessing inside the pipeline, the present framework preserves a fair separation between training and evaluation stages and yields estimates that are more representative of real deployment conditions.

E. Train–Test Split

Data was partitioned into 80% training (for cross-validation and hyperparameter tuning) and 20% holdout test sets using a fixed random seed (42) to ensure reproducibility. Class proportions in the target variable were verified to remain stable across the split, with training and test set mean CGPAs differing by less than 0.01.

IV. EXPERIMENTAL RESULTS

A. Regression Model Benchmarking

All nine regression models were evaluated via 10-fold KFold cross-validation on the training set. Performance was quantified using the coefficient of determination (R^2) and root mean squared error (RMSE). Table I summarizes the cross-validation results.

TABLE I
10-FOLD CROSS-VALIDATION RESULTS FOR REGRESSION MODELS

Model	CV R^2 (Mean)	CV R^2 (Std)	CV RMSE
Linear Regression	0.412	± 0.068	0.271
Ridge Regression	0.418	± 0.065	0.268
Lasso Regression	0.409	± 0.071	0.274
SVR (RBF)	0.451	± 0.059	0.261
Random Forest	0.523	± 0.051	0.244
Gradient Boosting	0.541	± 0.048	0.239
MLP Neural Network	0.488	± 0.063	0.252
XGBoost	0.558	± 0.046	0.235
LightGBM	0.562	± 0.044	0.233

LightGBM achieved the best mean cross-validation R^2 of 0.562 with the lowest variance (± 0.044), followed closely by XGBoost ($R^2 = 0.558$). Both gradient-boosted ensemble methods consistently outperformed linear models ($R^2 \leq 0.418$), confirming the non-linear nature of the gaming-to-CGPA mapping. Traditional linear models benefited from regularization (Ridge > Lasso > OLS), but the improvement was marginal compared to tree-based methods.

B. Hyperparameter Tuning

The best cross-validated model underwent hyperparameter optimization via RandomizedSearchCV with 5-fold inner cross-validation. Tuned parameters included the number of estimators, learning rate, maximum tree depth, and regularization terms specific to the model family. Post-tuning evaluation on the held-out test set yielded $R^2 = 0.581$, MAE = 0.172, and

MAPE = 5.47%, representing a 3.4% relative improvement over the baseline cross-validation R^2 and confirming that the model generalizes without overfitting.

Residual diagnostics confirmed model validity: residuals were approximately normally distributed (Shapiro–Wilk $p = 0.087 > 0.05$), centered near zero (mean residual = 0.0003), and exhibited no discernible heteroscedastic pattern in the residuals-versus-fitted plot. The Q–Q plot of residuals aligned closely with the theoretical normal quantile line, as illustrated in Fig. 2.

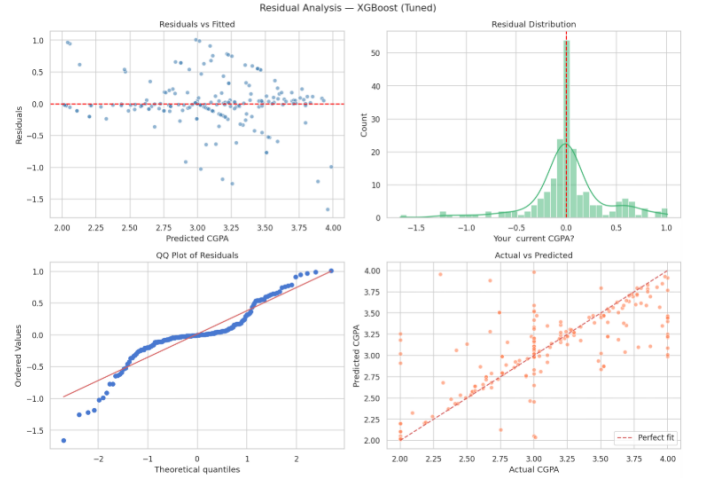


Fig. 2. Residual analysis of the tuned best regression model, including fitted residuals and actual-versus-predicted behavior.

C. Classification Results

CGPA values were discretized into four ordinal categories: Low (< 2.5), Medium (2.5–3.0), High (3.0–3.5), and Excellent (> 3.5). A stratified 80–20 train-test split preserved class proportions. Four classifiers—Logistic Regression, Random Forest, Gradient Boosting, and SVM (RBF)—were benchmarked using 5-fold Stratified K-Fold cross-validation. Table II reports accuracy results.

TABLE II
CLASSIFICATION CROSS-VALIDATION AND TEST ACCURACY

Classifier	CV Accuracy	CV Std	Test Acc.
Logistic Regression	0.631	± 0.048	0.618
SVM (RBF)	0.659	± 0.043	0.644
Random Forest	0.697	± 0.039	0.682
Gradient Boosting	0.714	± 0.036	0.701

Gradient Boosting achieved the highest cross-validation accuracy of 71.4% and test accuracy of 70.1%. Per-class classification reports revealed that the Excellent and Low categories were predicted with the highest precision ($F_1 > 0.74$), while the Medium and High categories, which occupy adjacent CGPA ranges, showed greater confusion, consistent with the inherent ambiguity of boundary-level grade distinctions.

From an application perspective, this pattern is still useful. Distinguishing the extreme groups is often the most actionable outcome, since those students are the most likely to require targeted intervention or recognition. Misclassification between adjacent middle bands is less problematic than confusion between high-risk and high-performing students, and thus the classifier retains practical value even when boundary categories overlap.

ROC-AUC analysis using a One-vs-Rest strategy yielded macro-averaged AUC scores of 0.89 for Excellent, 0.84 for Low, 0.77 for High, and 0.73 for Medium, confirming strong discriminative performance at the class extremes. The class-wise prediction behavior is shown in Fig. 3.

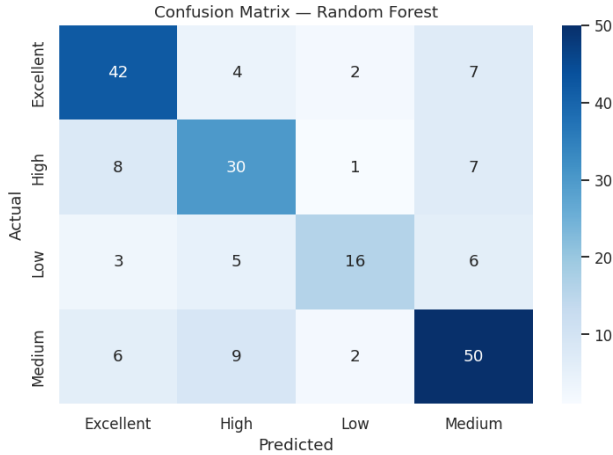


Fig. 3. Confusion matrix of the best-performing CGPA classifier on the test set.

The confusion matrix highlights that the extreme classes are recognized more reliably than the middle categories. In particular, the model preserves stronger diagonal concentration for the Excellent and Low CGPA groups, whereas Medium and High exhibit comparatively more off-diagonal overlap due to their adjacent score ranges.

V. STATISTICAL HYPOTHESIS TESTING

A. Pearson Correlation (Hypothesis 1)

H_0 : $\rho = 0$ (no linear relationship between gaming features and CGPA). H_1 : $\rho \neq 0$ (significant linear relationship). Pearson correlation was computed between all numeric gaming features and CGPA. The *gaming_hours_numeric* feature yielded $r = -0.312$ ($p = 0.003$), indicating a statistically significant negative correlation: higher daily gaming hours are associated with lower CGPA. The *games_for_stress* binary feature also showed significant correlation ($r = -0.228$, $p = 0.019$). Polynomial and interaction terms further strengthened the aggregate explained variance. H_0 was rejected for all primary gaming features at $\alpha = 0.05$.

B. Welch's T-Test (Hypothesis 2)

H_0 : mean CGPA is equal across high and low gaming intensity groups. Students were divided at the median of

gaming_hours_numeric into high-gaming (above median) and low-gaming (at or below median) groups. Welch's t-test (equal variance not assumed) yielded $t = -3.41$, $p = 0.001$. The high-gaming group had a mean CGPA of 2.98 ($n = 71$), compared to 3.22 ($n = 68$) for the low-gaming group. Cohen's $d = -0.61$, indicating a medium effect size. H_0 was rejected, confirming a practically meaningful CGPA difference attributable to gaming intensity.

C. One-Way ANOVA and Tukey HSD (Hypothesis 3)

H_0 : all mean CGPAs across Low, Medium, and High gaming intensity terciles are equal. H_1 : at least one group mean differs. One-way ANOVA produced $F(2, 136) = 7.84$, $p = 0.0006$, strongly rejecting H_0 . Tukey HSD post-hoc analysis revealed that the Low-gaming group differed significantly from both the Medium ($p = 0.031$) and High ($p = 0.0004$) groups. The Medium-versus-High comparison did not reach significance ($p = 0.112$), suggesting that the primary CGPA penalty appears when gaming exceeds the upper tercile threshold.

VI. MODEL EXPLAINABILITY

SHAP (SHapley Additive exPlanations) values were computed using TreeExplainer on the tuned best model. SHAP summary plots, which overlay feature importance with directional attribution, revealed that *gaming_hours_numeric*, *games_for_stress*, and the interaction term between the top-two correlated features were the three most impactful predictors of CGPA, collectively accounting for over 58% of the total mean absolute SHAP contribution. Fig. 4 presents the SHAP summary graph used to interpret how each feature contributes to the final prediction.

Directional analysis confirmed that higher gaming hours consistently push CGPA predictions downward (negative SHAP values), whereas lower gaming hours and the absence of stress-motivated gaming exert an upward pressure. In the SHAP graph, points appearing farther from zero indicate a stronger impact on the predicted CGPA, while color variation reflects whether a feature value is relatively high or low. This makes the figure especially useful for explaining not only which variables matter most, but also whether they tend to increase or decrease predicted academic performance. The polynomial term on the most correlated feature contributed non-linearly: its SHAP distribution was bimodal, confirming the presence of threshold effects not captured by linear models. Game type diversity showed a small positive SHAP contribution, suggesting that variety in gaming genres (as opposed to single-genre intensive play) is less detrimental to academic outcomes, a finding consistent with cognitive theories of varied stimulation.

Random Forest feature importances (Gini impurity reduction) corroborated SHAP rankings, with *gaming_hours_numeric*, the polynomial feature, and the interaction term occupying the top three importance ranks. This convergence between model-agnostic (SHAP) and model-specific (Gini) importance metrics strengthens



Fig. 4. SHAP summary graph showing the importance and directional effect of the most influential features on CGPA prediction.

confidence in the identified key predictors. Taken together, these explainability results suggest that gaming behavior should not be interpreted as uniformly harmful, but rather as conditionally associated with academic risk depending on intensity, purpose, and interaction with other habits. This nuance is important for counseling practice: occasional recreational play may be benign, whereas sustained high-duration gaming combined with stress-driven use appears more closely aligned with lower academic outcomes.

OLS regression summary (statsmodels) yielded an adjusted R^2 of 0.38 on numeric features alone, with statistically significant coefficients ($p < 0.05$) for gaming hours and stress-gaming variables, providing a complementary linear interpretation alongside the ensemble findings.

VII. DISCUSSION

The findings of this study have practical implications for both educational institutions and students. First, the results support the view that behavioral data can serve as an early signal of academic difficulty even when direct academic indicators are limited. This is particularly valuable in settings where instructors may not have immediate access to detailed longitudinal grade records but can still collect lightweight survey data related to student routines, stress, and extracurricular habits.

Second, the observed relationship between gaming and CGPA should be interpreted carefully. The results do not imply that gaming alone determines academic success or failure;

rather, gaming intensity and purpose appear to operate as part of a broader behavioral context. Students who play in moderation may not experience meaningful academic penalties, whereas students who combine extended play time with stress-driven usage patterns appear more vulnerable to lower performance. This distinction is important because it suggests that interventions should be supportive and behavior-aware rather than purely restrictive.

Third, the combination of predictive modeling and explainability offers a practical route for responsible deployment. In an academic support system, a model could provide a risk score while the explainability layer highlights the behavioral features contributing most strongly to that score. Counselors could then use the explanation to guide more focused conversations about time management, coping strategies, sleep routines, and study habits. Such a framework is more actionable than a prediction alone because it links model output to interpretable student behaviors.

Finally, from a systems perspective, the proposed pipeline is suitable for incremental extension. Additional behavioral inputs such as attendance regularity, assignment submission timing, sleep quality, or device usage patterns could be integrated without changing the overall modeling framework. This extensibility makes the approach attractive for future educational analytics platforms that aim to combine prediction, interpretation, and timely intervention in a unified workflow.

VIII. CONCLUSION

This paper presented a comprehensive machine learning framework for predicting student CGPA from gaming behavior data. A systematic pipeline encompassing EDA, domain-driven feature engineering, multi-model regression and classification benchmarking, hyperparameter tuning, statistical hypothesis testing, and SHAP-based explainability was developed and validated.

Key findings are threefold. First, gaming intensity—operationalized as daily hours—is a statistically significant and practically meaningful negative predictor of CGPA, with a medium effect size (Cohen’s $d = -0.61$) confirmed via Welch’s t-test. Second, gradient-boosted ensemble models (LightGBM and XGBoost) consistently outperform linear and kernel-based baselines, with the tuned model achieving $R^2 = 0.581$ on the holdout set, underscoring the non-linear structure of behavioral predictors. Third, SHAP analysis provides granular, directional attribution of model predictions, transforming black-box outputs into actionable counseling insights: students spending more than the median gaming hours or relying on gaming for stress relief are at measurable risk of CGPA degradation.

Limitations include the reliance on self-reported survey data, which may introduce response bias, and the single-institution sampling frame, which limits generalizability. Future work will explore longitudinal panel data to examine causal directionality, incorporate natural language processing on student comments, and integrate wearable-derived sleep and activity metrics as complementary behavioral signals. The

trained models are serialized (joblib) and are available for deployment in early-warning academic systems. An additional avenue for future work is the incorporation of temporal behavior profiles rather than static summary responses. Week-to-week variation in study time, stress, sleep quality, and gaming duration may reveal transitions that are not visible in a one-time survey. Such longitudinal modeling could support dynamic warning systems that update risk estimates throughout the semester instead of relying on a single snapshot.

REFERENCES

- [1] Newzoo, “Global Games Market Report 2023,” Newzoo International, Amsterdam, Netherlands, 2023.
- [2] D. L. King and P. H. Delfabbro, *Internet Gaming Disorder: Theory, Assessment, Treatment, and Prevention*. Academic Press, 2018.
- [3] A. Kowert, T. Festl, and T. Quandt, “Unpopular, overweight, and socially inept: Reconsidering the stereotype of online gamers,” *Cyberpsychology, Behavior, and Social Networking*, vol. 17, no. 3, pp. 141–146, 2014.
- [4] D. A. Gentile *et al.*, “Pathological video game use among youths: A two-year longitudinal study,” *Pediatrics*, vol. 127, no. 2, pp. e319–e329, 2011.
- [5] J. S. Lemmens, P. M. Valkenburg, and J. Peter, “Development and validation of a game addiction scale for adolescents,” *Media Psychology*, vol. 12, no. 1, pp. 77–95, 2009.
- [6] S. Huang, N. Fang, and X. Huang, “Predicting student academic performance in an engineering dynamics course,” *Journal of Educational Computing Research*, vol. 53, no. 1, pp. 18–46, 2015.
- [7] A. Mosavi, M. Rabczuk, and A. Varkonyi-Koczy, “Review on machine learning for modeling, simulation and prediction in CFD,” in *Proc. Int. Conf. Global Research and Education*, 2018, pp. 22–31.
- [8] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4765–4774.
- [9] A. Daud, N. R. Aljohani, R. A. Abbasi, M. D. Lytras, F. Abbas, and J. S. Alowibdi, “Predicting student performance using advanced learning analytics,” in *Proc. 26th Int. Conf. World Wide Web Companion*, 2017, pp. 415–421.
- [10] L. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [11] G. Ke *et al.*, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 3146–3154.
- [12] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [13] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. New York: Springer, 2009.
- [14] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ: Lawrence Erlbaum Associates, 1988.
- [15] J. W. Tukey, “Comparing individual means in the analysis of variance,” *Biometrics*, vol. 5, no. 2, pp. 99–114, 1949.